



Holistic reaction template generation using context-driven approach

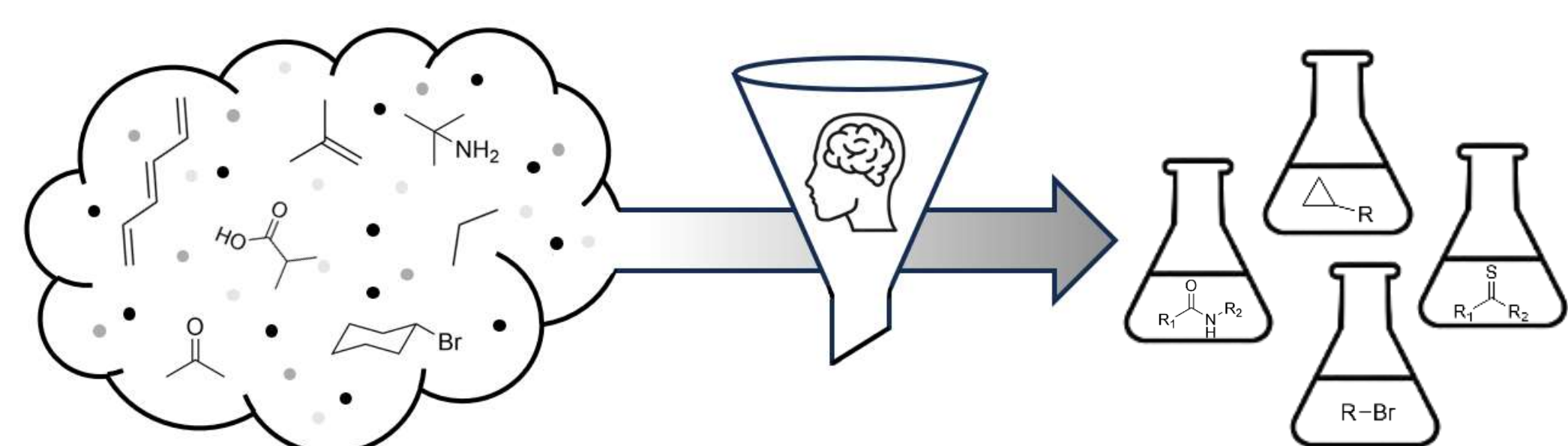
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Introduction

Abstract

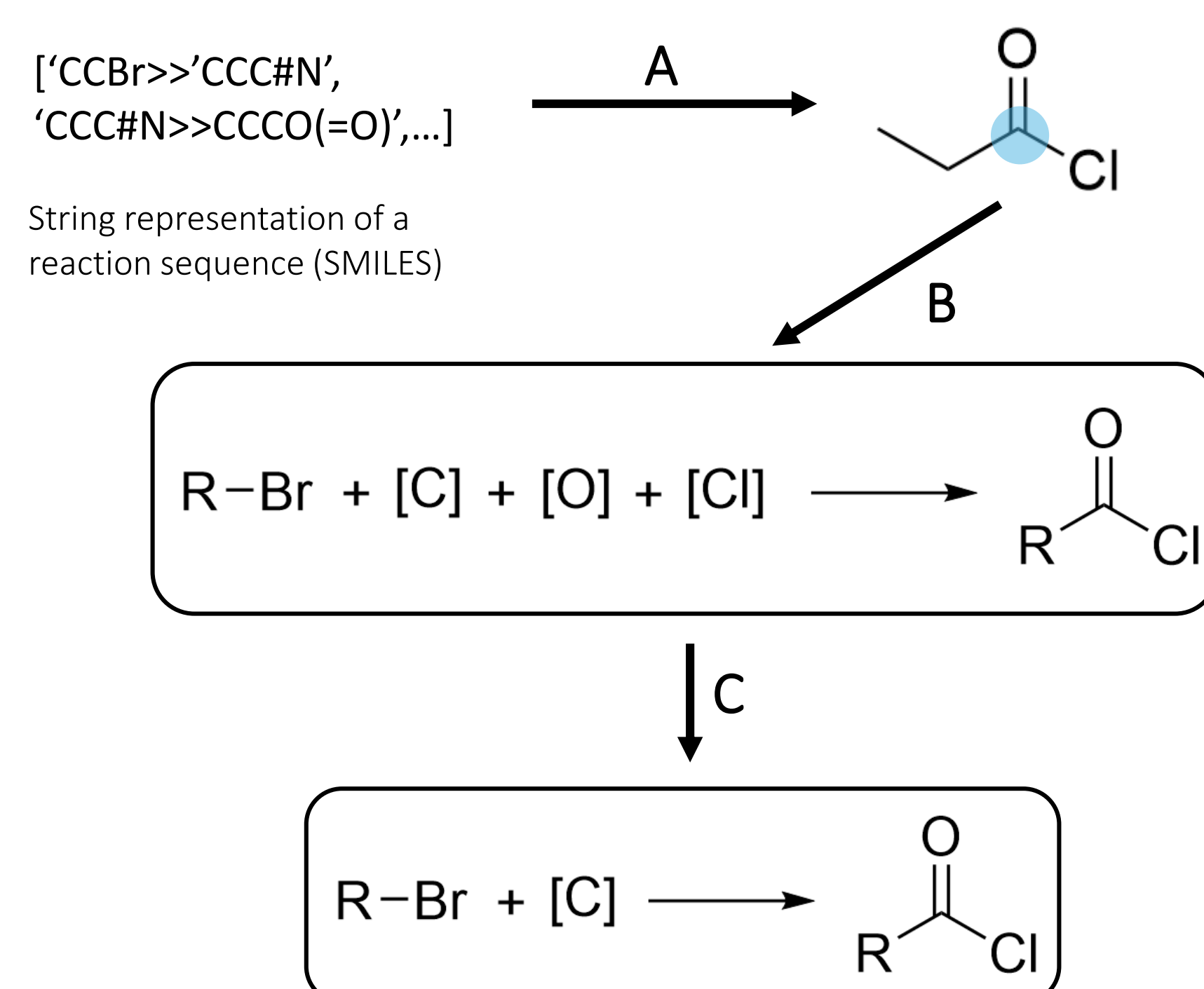
The development of novel organic materials and medicine hinges on the creation and refinement of innovative synthetic routes. So how can we narrow this vast chemical space using human intuition to guide us in determining an impactful class of reactions?



The key step is to identify recurring candidate reaction patterns/sequences among the millions of syntheses in literature and design an ideal reaction based on their framework. This not only cuts down on the multistep nature of synthesis, it also allows for the synthetic scheme to produce higher yield at a lower cost.

Overview

The overview of our approach to analyzing a reaction patterns is given below:



A: Context-based reaction center extraction

From the SMILES of a synthetic route, pull contextual reaction centers

B: Granular reaction template extraction

From reaction centers, generate granular templates

C: Granularity reduction

Using the granular template, reduce specificity through chemical intuition

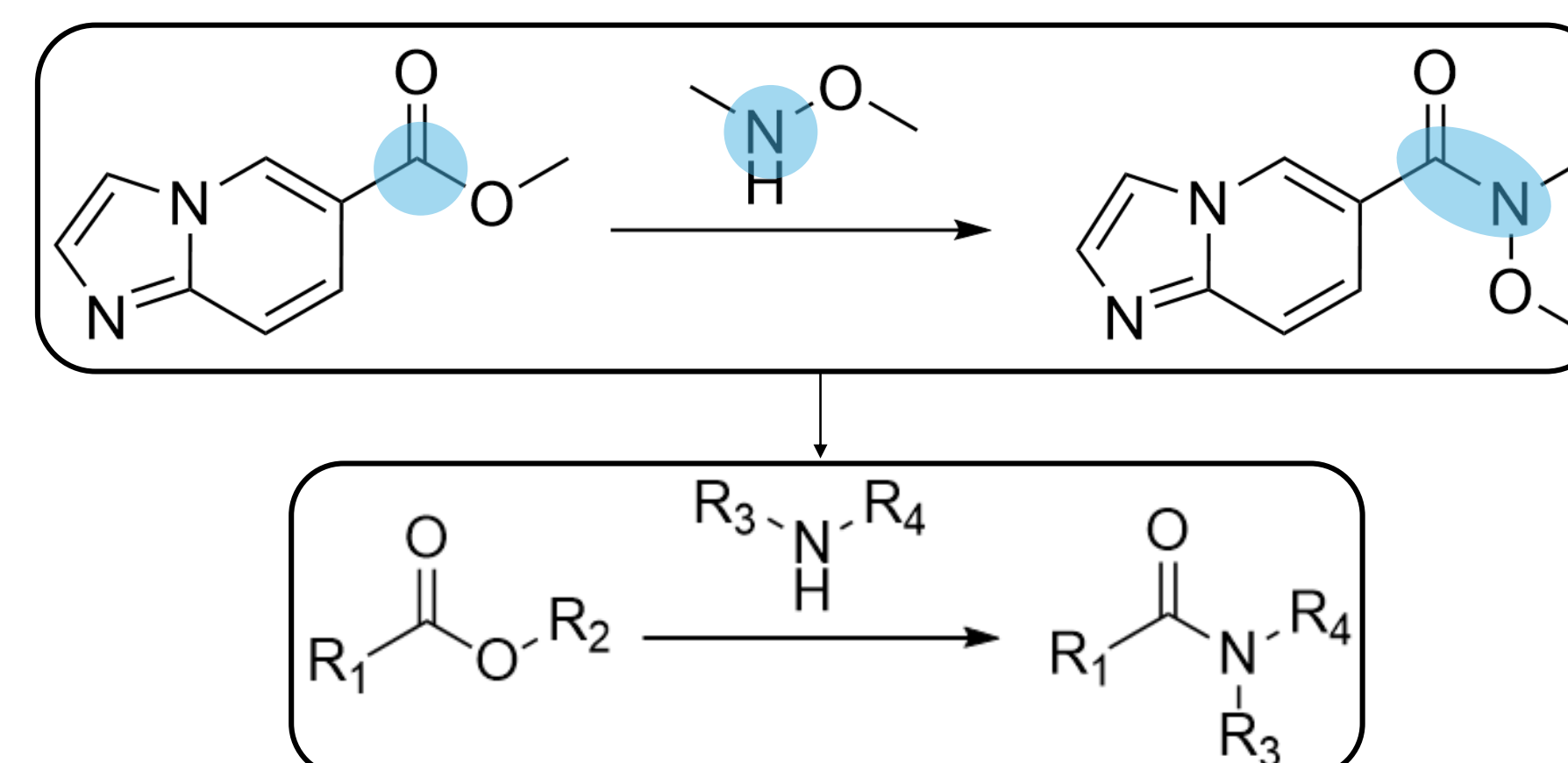
Methods

Previous Work

- Atomistic Reaction Centers

An atomistic centers are atoms within the reactant molecules that experience bonding changes in one reaction step.

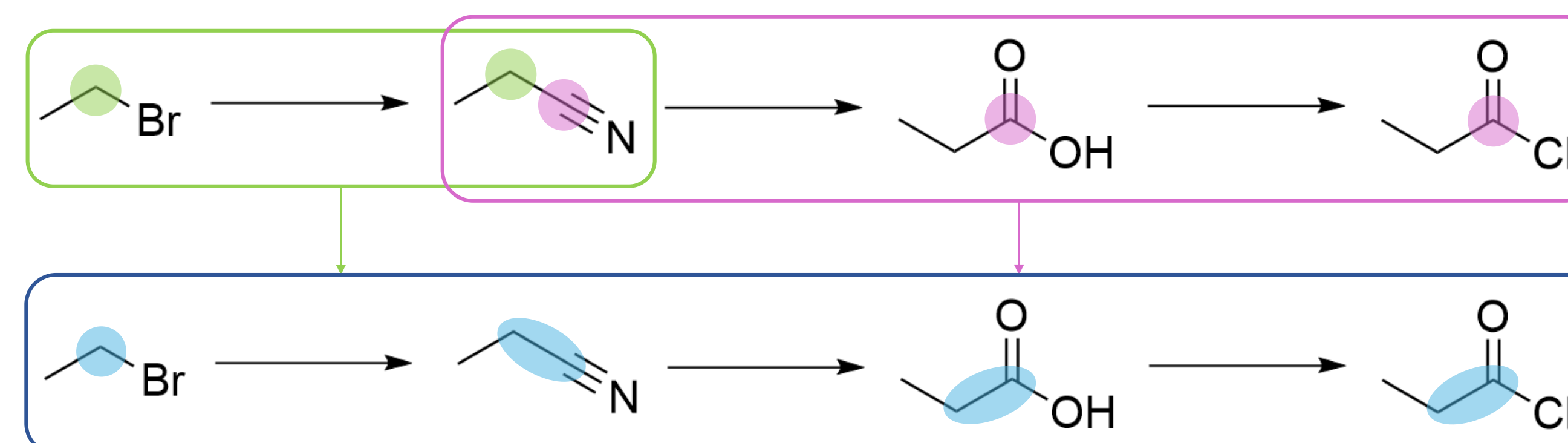
An atomistic template generator (Coley et al.) will extract the local template by determining this atomistic center and returning the transformations around the center.



This Work

- Context-based reaction centers

With the context of a synthetic route, we extract the context-based reaction center from each reaction's atomistic centers

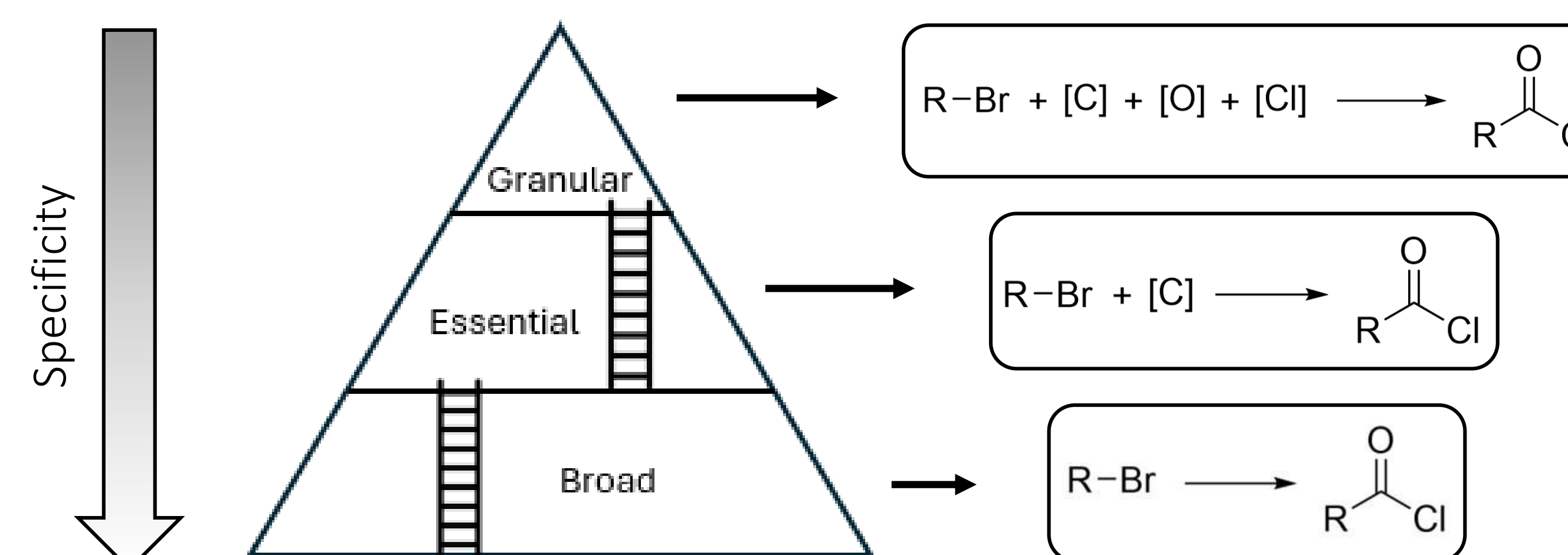


To extract them, we perform a connectivity check based on chemical intuition across all atomistic reaction centers and group them, within the context of the synthesis

- Granular template generation and reduction:

Using the context-based reaction centers obtained in the previous step, we generate a granular reaction template with every necessary but general info.

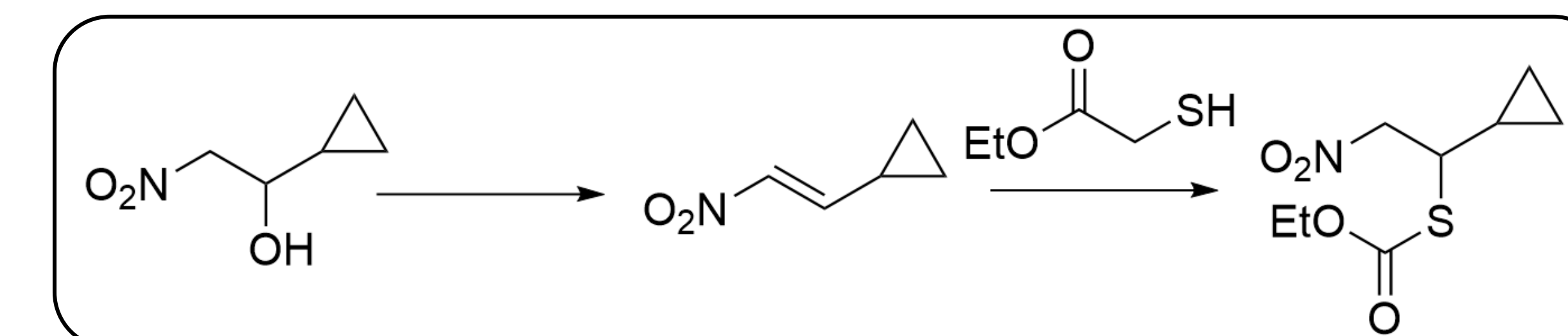
To modulate the granularity of the template, we systematically choose and remove redundant information (and climb down the ladder of granularity)



Results

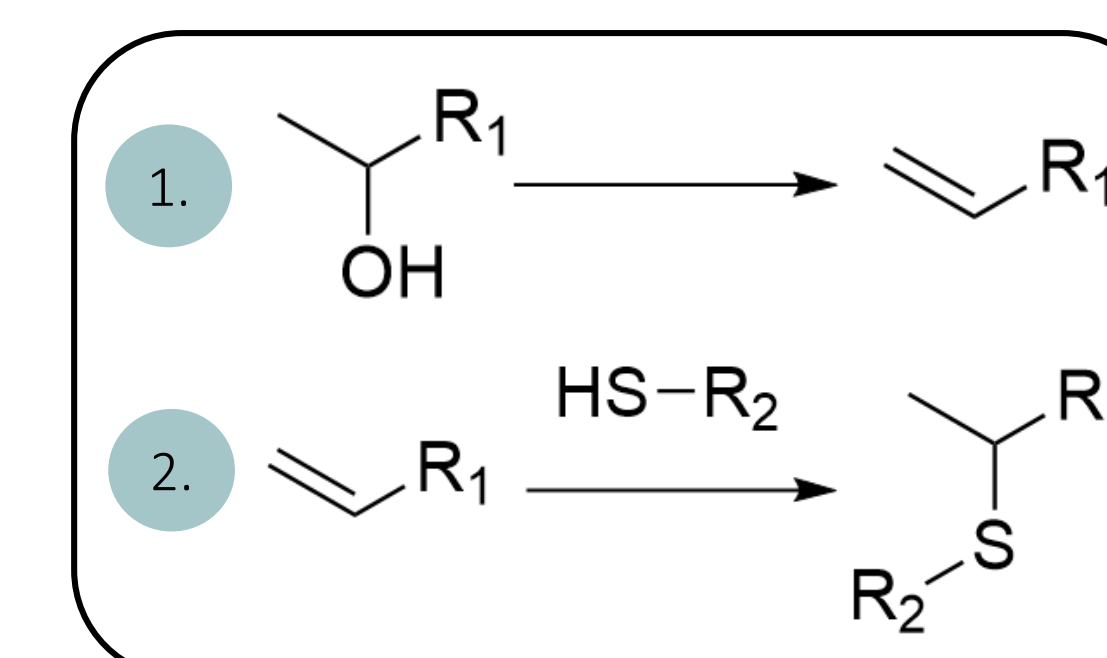
Comparison

- Context-based approach vs Atomistic approach



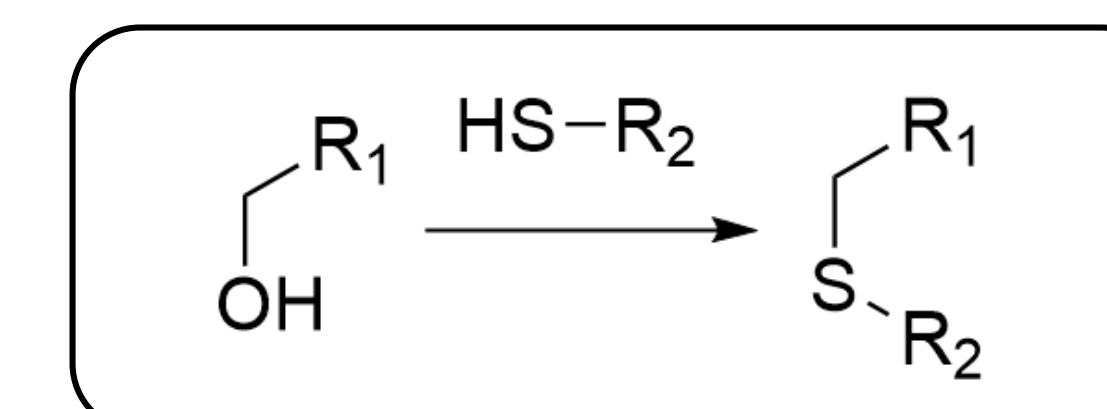
- Previous work:

Atomistic reaction centers generates templates for dehydration and thiol addition, but misses the overall alcohol to thiol functionalization



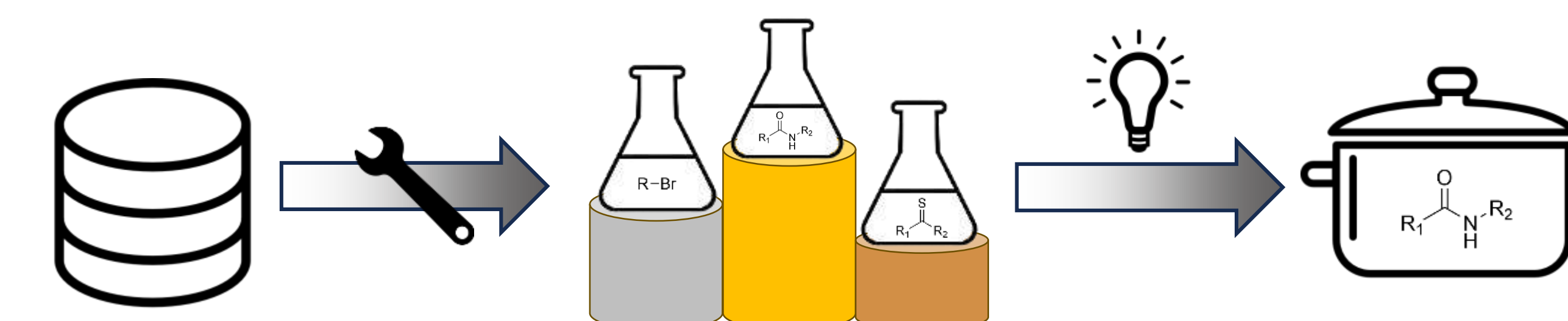
- This work:

Context-based reaction centers correctly identifies the functionalization and returns the corresponding template



Future Work

- Holistic template generate on a US patent database of millions of synthesis to elucidate the most common and impactful sequence of reactions.
- Experimentally design one step reaction procedures to accomplish the holistic templates generated.



References

- Connor W. Coley, William H. Green, Klavs F. Jensen (2019) RDChiral: An RDKit Wrapper for Handling Stereochemistry in Retrosynthetic Template Extraction and Application.